

First transatlantic
cable: pulses smear

T1: 1.544 Mb/s
AMI on copper

Ethernet at PARC
(Manchester)

PRML disk
read channels

10GbE: 64b/66b
+ scrambling

PCIe 6.0:
PAM-4 flits

1928

1963

1983

1995

2017

2026

1858

1962

1973

1990

2002

2022

Nyquist: $2W$ pulses/s
without interference

Lender:
duobinary

Widmer & Franaszek:
8b/10b

Fast Ethernet:
4B5B + MLT-3

400GbE:
PAM-4 + FEC

200 Gb/s
per lane